

DATABASE SYSTEMS

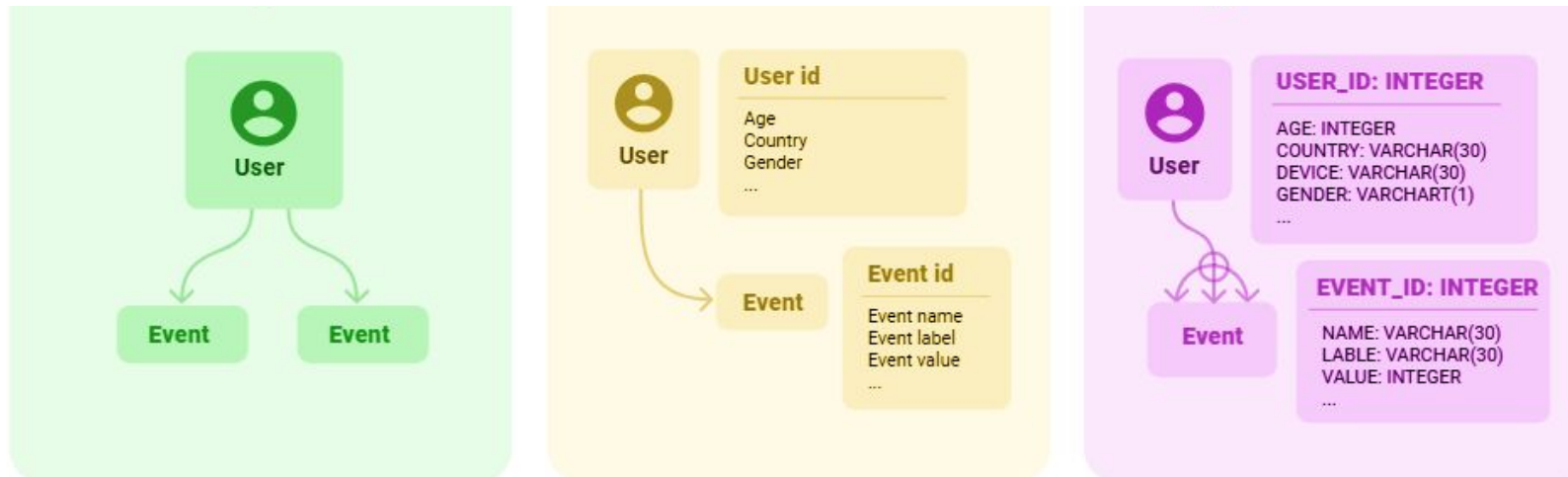
WEEK-05

Data Models



DATA MODELS

- Database model shows the logical structure of a database, including the relationships and constraints that determine how data can be stored and accessed.
- Most data models can be represented by an accompanying database diagram.
- Therefore, there are following four data models used for understanding the structure of the database:



DATA MODELS AND CONCEPTUAL MODELING

- Data Model is higher level of description of schema
- Describes, manipulate, relationship & constraints on data
- Three parts of data Model:
 - ❑ Structural part: Set of rules to construct database
 - ❑ Manipulative part: Set of operations
 - ❑ Set of integrity constraints: Ensures data is accurate



CONCEPTUAL DATA MODELS

- Conceptual data models provide a high-level view of data by identifying the key entities, attributes, and relationships.
- For example, a conceptual data model for a library may include entities like 'Book,' 'Author,' and 'Publisher' and their relationships.
- These models are often used in the initial planning phase to define the broad data structure without going into detailed attributes.



LOGICAL DATA MODELS

- Logical data models offer a detailed overview of the data without any concern for how the data will be physically implemented in databases.
- They include all entities, attributes, and relationships, along with the addition of primary and foreign keys.
- They also define data types for attributes.
- In the library example, a logical data model would specify the data type for 'Book Title' as a string and 'Publication Date' as a date.



PHYSICAL DATA MODELS

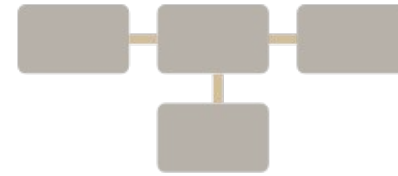
- Physical data models are the most detailed and are tailored to a specific database management system (DBMS).
- They include all tables, columns, relationships, indexes, triggers, and stored procedures.
- They specify how data will be stored in a database, considering the performance requirements and the specific technology used.
- In the library example, a physical data model would detail how the 'Book' table is physically stored in the database.



DATA MODELS



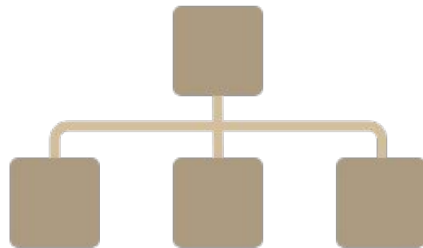
Entity relationship



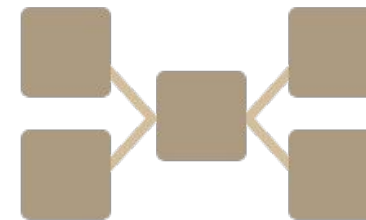
Object oriented



Relational



Hierarchical



Dimensional

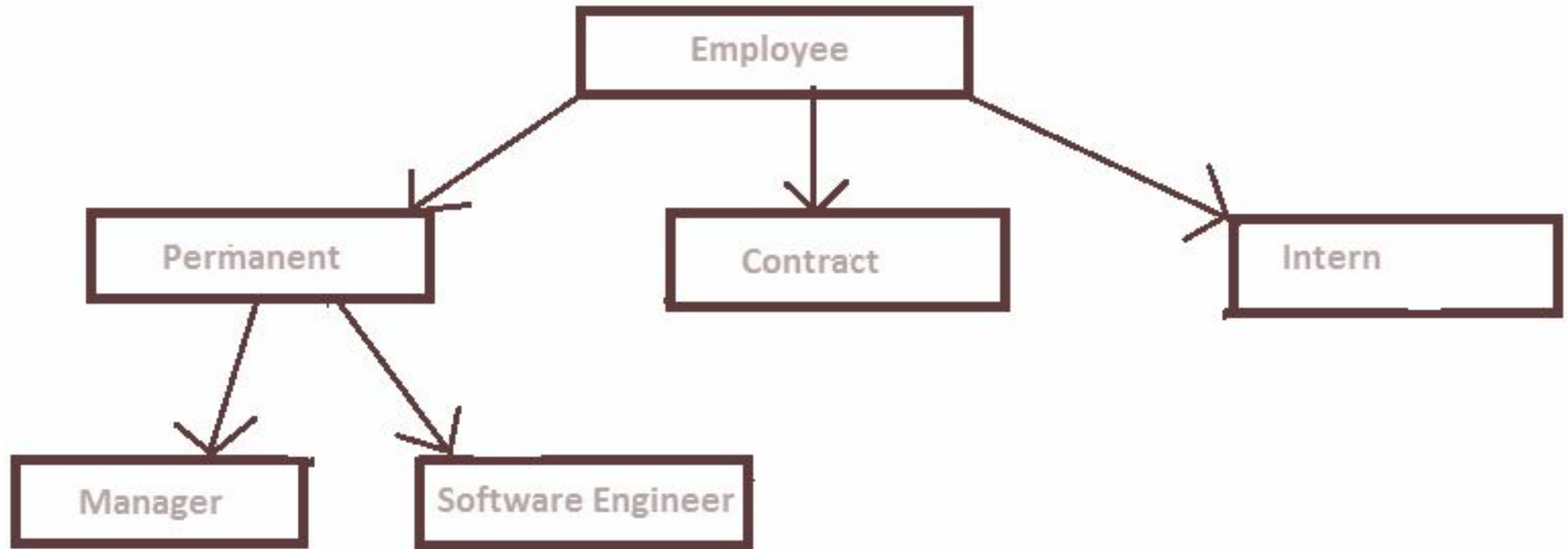


HIERARCHICAL MODEL

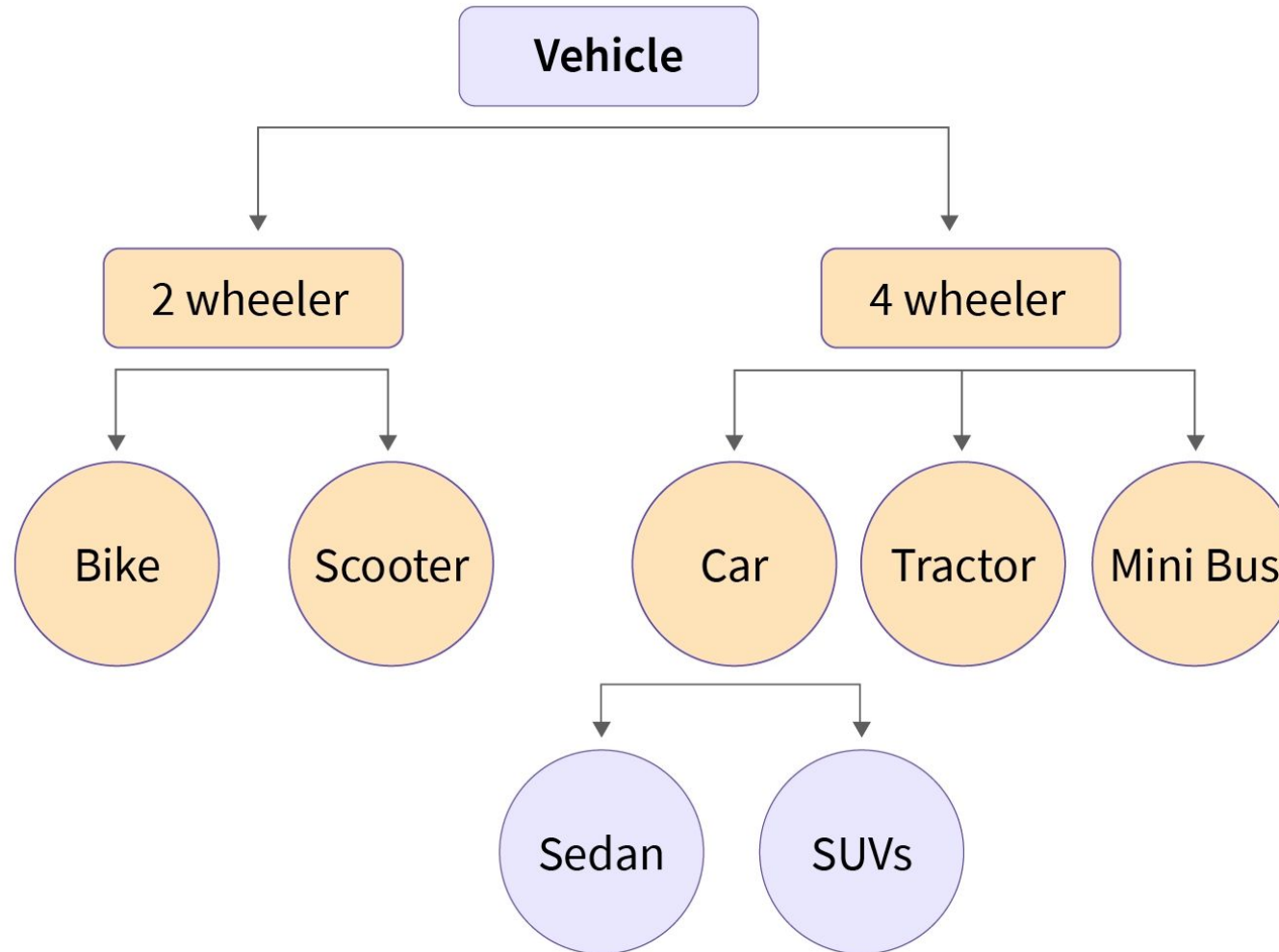
- The model data are represented by collection of records
- The hierarchical database model organizes data into a **tree-like structure**, with a **single root**, to which all the other data is linked.
- The hierarchy starts from the **Root** data, and expands like a tree, adding **child** nodes to the **parent** nodes.
- In this model, a child node will only have a **single parent node**.
- Data is organized into a tree-like structure with a **one-to-many relationship** between two different types of data
- IBM's Information Management System (IMS) is based on this model
- Example, one **department** can have many **courses**, many **teachers**, and of course many **students**.



HIERARCHICAL MODEL



HIERARCHICAL MODEL



HIERARCHICAL MODEL

Advantages –

- Simplicity, Data Integrity, Data security, Efficiency, Easy availability of expertise.

Disadvantages –

- Complexity, Inflexibility, Lack of Data Independence,
- But the Hierarchical model is less flexible and it doesn't support many-to-many relationships.



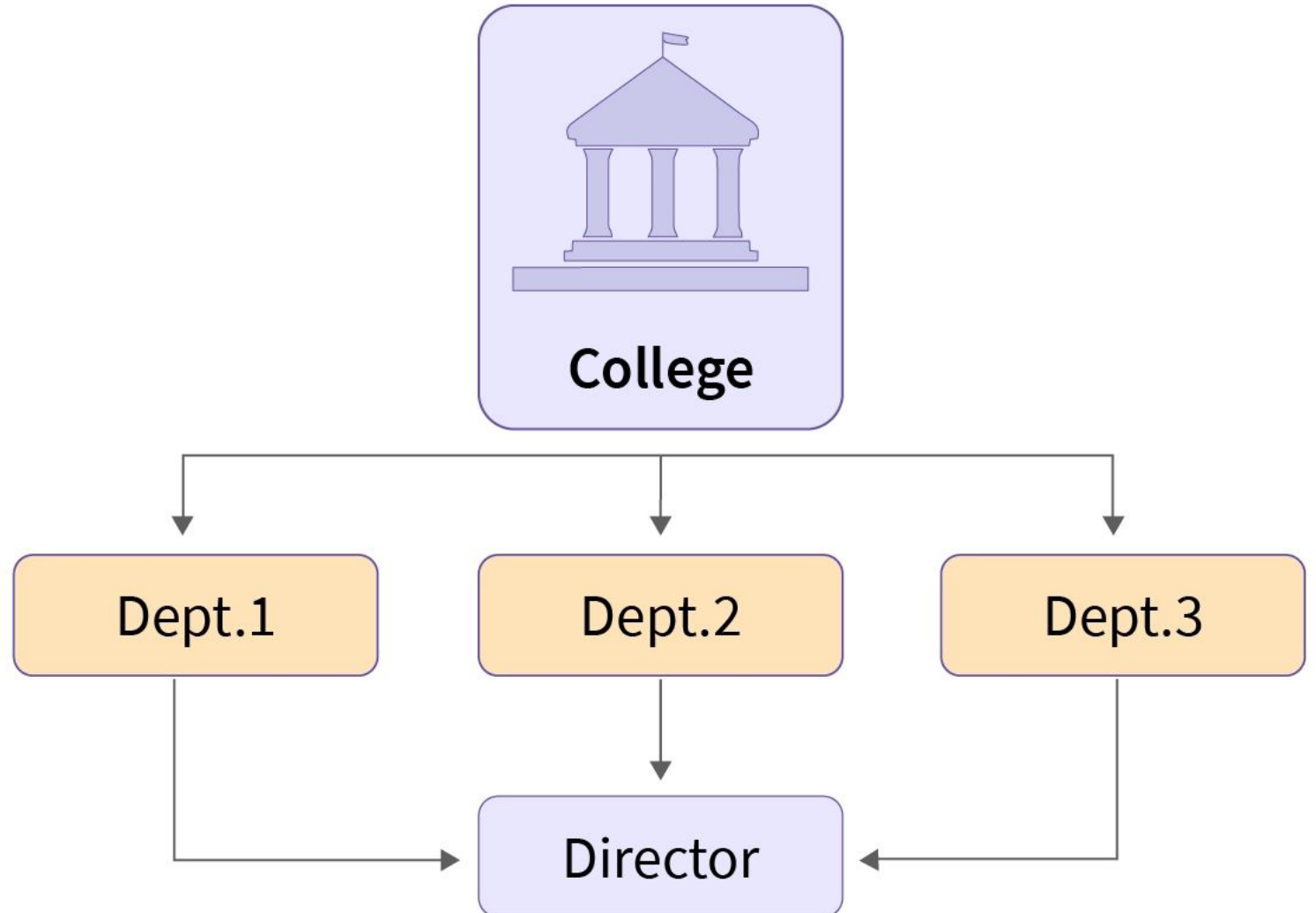
NETWORK DATA MODEL

- This model data are represented by collection of records
- Relationships among the data are represented by links
- Graph data structures are used in this model
- It permits a record to have more than one parent
- This database model uses many-to-many data relationships.
- For Example- Social Media sites like Facebook, Instagram etc
- **Advantages –**
Simplicity, Data Integrity, Data Independence, Database standards.
- **Disadvantages –**
System Complexity, Lack of structural Independence.

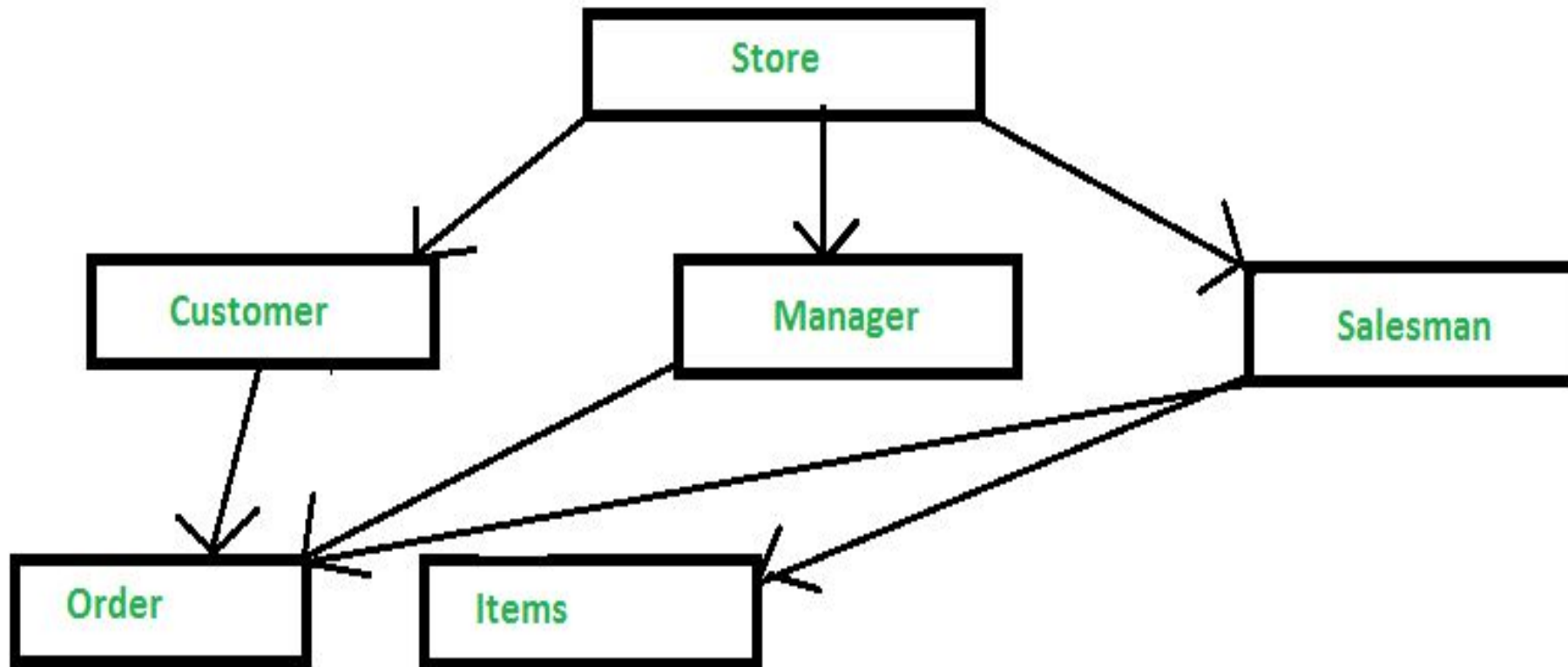


NETWORK DATA MODEL

- All three departments are linked with the director which was not possible in the hierarchical data model.



NETWORK DATA MODEL



Entity-Relationship Data Model

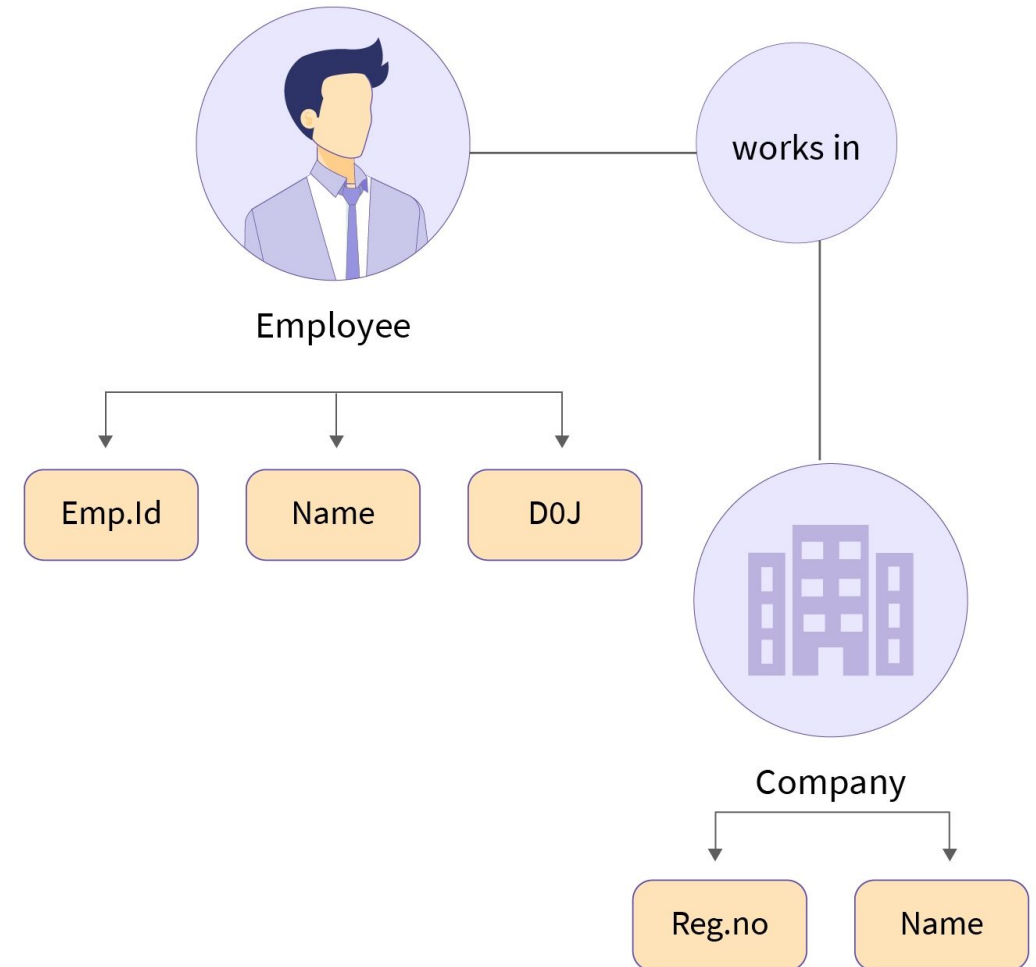
- Designed by Peter Chen and published in 1976 papers.
- An ER model is the logical representation of data as objects and relationships among them.
- These objects are known as entities, and relationship is an association among these entities.
- A set of attributes describe the entities.
- For example, student_name, student_id describes the 'student' entity.
- A set of the same type of entities is known as an 'Entity set', and the set of the same type of relationships is known as 'relationship set'.
- It was widely used in database designing.



ENTITY-RELATIONSHIP DATA MODEL

Two entities that are Employee and Company, and the relationship among them.

Both the employee and company have some attributes and the relationship is of "works in" type, which means the employee works in a company.



Object-based Data Model

- An extension of the ER model with notions of functions, encapsulation, and object identity, as well.
- This model supports a rich type system that includes structured and collection types.
- A multimedia database incorporates media, such as images, that could not be stored in a relational database.
- A hypertext database allows any object to link to any other object.
- Thus, in 1980s, various database systems following the object-oriented approach were developed. Here, the objects are nothing but the data carrying its properties.
- A very popular example of an Object Database management system or **ODBMS** is **MongoDB** which is also a NoSQL database.



Semi structured Data Model

- This type of data model is different from the other three data models.
- The semi structured data model allows the data specifications at places where the individual data items of the same type may have different attributes sets.
- The Extensible Markup Language, also known as XML, is widely used for representing the semi structured data.
- Although XML was initially designed for including the markup information to the text document, it gains importance because of its application in the exchange of data.

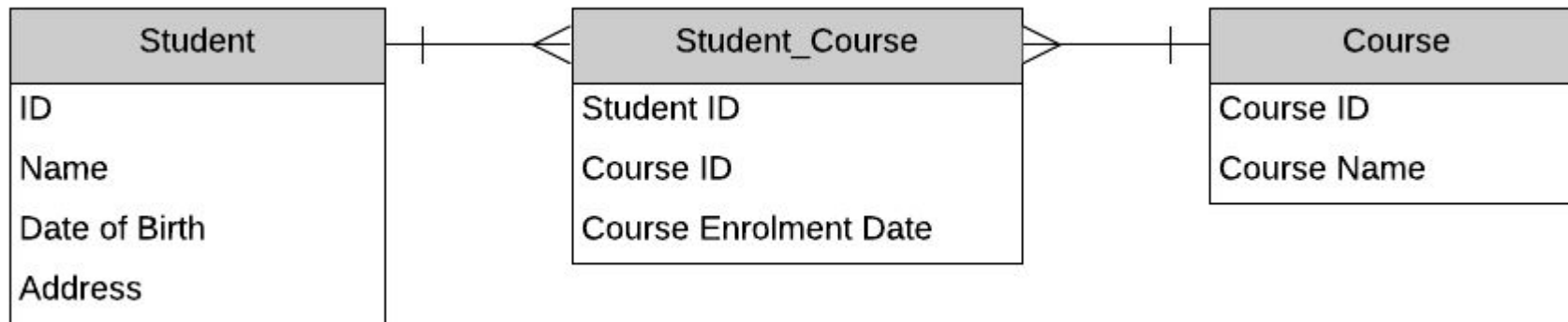


Relational Data Model

- Initially described by Edgar F. Codd, in 1969.
- This type of model designs the data in the form of rows and columns within a table.
- Thus, a relational model uses tables for representing data and in-between relationships.
- Tables are also called relations.
- The relational data model is the widely used model which is primarily used by commercial data processing applications.



RELATIONAL DATA MODEL



RELATIONAL DATA MODEL

Activity Code	Activity Name
23	Patching
24	Overlay
25	Crack Sealing

Key=24

Activity Code	Date	Route No.
24	01/12/01	I-95
24	02/08/01	I-66

Date	Activity Code	Route No.
01/12/01	24	I-95
01/15/01	23	I-495
02/08/01	24	I-66

Stu. Id	Name	Branch
101	Naman	CSE
102	Saloni	ECE
103	Rishabh	IT
104	Pulkit	ME



RDBMS (RELATIONAL DATABASE MANAGEMENT SYSTEM)

- All modern database management systems like SQL, MS SQL Server, IBM DB2, ORACLE, My-SQL, and Microsoft Access are based on RDBMS.

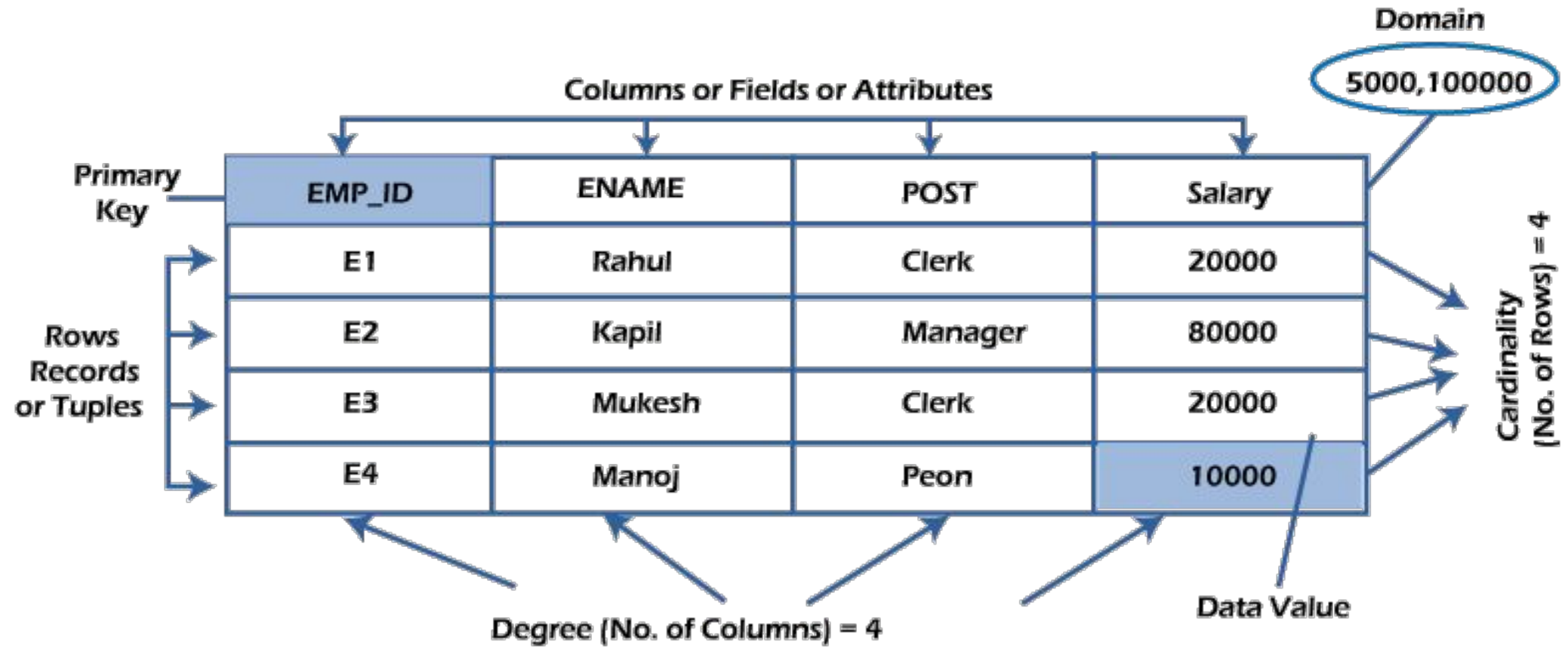


HOW IT WORKS

- Data is represented in terms of tuples (rows) in RDBMS.
- A relational database is the most commonly used database. It contains several tables, and each table has its primary key.
- Due to a collection of an organized set of tables, data can be accessed easily in RDBMS.



TERMINOLOGIES OF RDBMS



WHAT IS TABLE/RELATION?

- Everything in a relational database is stored in the form of relations.
- The RDBMS database uses tables to store data.
- A table is a collection of related data entries and contains rows and columns to store data.



EXAMPLE OF TABLE

ID	Name	AGE	COURSE
1	Ajeet	24	B.Tech
2	aryan	20	C.A
3	Mahesh	21	BCA
4	Ratan	22	MCA
5	Vimal	26	BSC



WHAT IS A ROW OR RECORD?

- A row of a table is also called a record or tuple.
- It contains the specific information of each entry in the table.
- It is a horizontal entity in the table. For example, The above table contains 5 records.

ID	Name	AGE	COURSE
1	Ajeet	24	B.Tech



What is a column/attribute?

- A column is a vertical entity in the table which contains all information associated with a specific field in a table.
- For example, "name" is a column in the above table which contains all information about a student's name.



Properties of an Attribute:

- Every attribute of a relation must have a name.
- Null values are permitted for the attributes.
- Default values can be specified for an attribute automatically inserted if no other value is specified for an attribute.
- Attributes that uniquely identify each tuple of a relation are the primary key.



Name

Ajeet

Aryan

Mahesh

Ratan



What is data item/Cells?

- The smallest unit of data in the table is the individual data item. It is stored at the intersection of tuples and attributes.
- Properties of data items:
- Data items are atomic.
- The data items for an attribute should be drawn from the same domain.

ID	Name	AGE	COURSE	ID
1	Ajeet	24	B.Tech	1



Degree:

- The total number of attributes that comprise a relation is known as the degree of the table.

For example, the student table has 4 attributes, and its degree is 4.

ID	Name	AGE	COURSE
1	Ajeet	24	B.Tech
2	aryan	20	C.A
3	Mahesh	21	BCA
4	Ratan	22	MCA

